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1 AI Agent Designer - Build Your First Agent

SkillForge Virtual Internship

AI Agent Designer - Build Your First Agent

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python / Jupyter Notebooks, ChatGPT / Claude (Prompting), LangGraph / LangChain, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **NeuralFrontier**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from NeuralFrontier

NeuralFrontier is a leading global enterprise operating at the cutting edge of the Design UI/UX sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'.

With an annual revenue exceeding 5.4billionandapresenceinover30countries, we collectivelyemployover42,000passionateprofessionals.

This job simulation is with our advanced innovation division, NeuralFrontier InsightsGen – erative Technologies. You will gain insight into how our multi – disciplinary expertssolvesomeofthe most complex business problems using state – of – the – art AI solutions.

The Problem:

As an AI Agent Designer, you are tasked with creating an intelligent virtual assistant capable of helping users navigate complex tasks efficiently. From understanding user goals to integrating memory and evaluating performance, your mission is to design an AI agent that balances functionality, usability, and ethical considerations. Imagine working on a groundbreaking AI product for a global tech company that aims to revolutionize customer service, productivity, and personal assistance.

Your mission is to design, build, and test an AI agent equipped with conversational capabilities, memory retention, and a goal-oriented problem-solving approach. The agent should be able to understand user intent, recall relevant information, and deliver solutions effectively. You will also focus on responsible AI practices, ensuring that your agent meets ethical standards while offering high usability.

What Will You Learn?

- Analyze Design UI/UX workflows to identify Agentic AI opportunities
- Design multi-agent orchestration and human-in-the-loop handoffs
- Implement conversational AI state machines
- Validate outputs for hallucination and safety

What Tools Will You Learn?

- Python / Jupyter Notebooks
- ChatGPT / Claude (Prompting)
- LangGraph / LangChain
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand

Scenario: You are tasked with designing an AI agent for a productivity app. The agent will assist users in scheduling meetings, tracking tasks, and answering queries. **Student Assignment:** Research and define the core functionalities of the agent. Create a user persona and list potential use cases for the agent. **Deliverable:** - A document outlining the agent's purpose, target audience, and core functionalities. - A list of 5-7 use cases based on the user persona.

Task 2: Data/Setup

Scenario: To enable memory and goal-oriented behavior, you need to set up the foundational architecture of the AI agent. This includes preparing the environment and connecting to the OpenAI API. **Student Assignment:** Set up Python, install necessary libraries (OpenAI, LangChain), and connect to the GPT model. Design a basic agent skeleton with input/output flow. **Deliverable:** - Python script with API setup and a basic agent scaffold. - Screenshot of the agent successfully generating sample responses.

Task 3: Build/Execute

Scenario: You will now integrate memory functionality so the agent can retain and recall user information across sessions. Add goal-setting logic to allow the agent to guide users toward task completion. **Student Assignment:** Use LangChain to implement memory and goal-setting modules. Train the agent to store user preferences and recall them in future interactions. Test the agent's ability to track goals across conversations. **Deliverable:** - Python script with memory integration and goal-setting logic. - A set of test cases demonstrating that the agent can recall memory and guide users to complete tasks.

Task 4: GenAI/Explanation

Scenario: Users often need step-by-step instructions for complex processes. Add a generative layer to the agent for explaining multi-step tasks clearly. **Student Assignment:** Integrate OpenAI's GPT capabilities to enable the agent to generate step-by-step instructions for tasks such as scheduling a meeting or organizing a to-do list. **Deliverable:** - Python script demonstrating generative capabilities for step-by-step instructions. - A test log showing the agent successfully explaining three distinct workflows.

Task 5: Audit/Responsible AI

Scenario: You must ensure the agent adheres to responsible AI practices. This includes avoiding biased responses, ensuring user data privacy, and providing disclaimers for ambiguous answers. **Student Assignment:** Implement filters to detect and manage biased or inappropriate responses. Add user consent mechanisms for data storage and integrate disclaimers for uncertain outputs. **Deliverable:** - Python script with responsible AI features implemented. - A document outlining responsible AI measures and testing results.

Task 6: Present Recommendation

Scenario: Your agent is ready for deployment, but you need to present its capabilities to the product team. Create a demo showcasing the agent's functionalities and explain its design choices. **Student Assignment:** Develop a short presentation and record a demo of the agent performing key tasks, showcasing its memory, goal-setting, and instructional capabilities. Highlight responsible AI features. **Deliverable:** - Presentation slides (5-10 slides) covering agent design, features, and responsible AI considerations.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

2 Microfinance Analyst - AI Loan Eligibility Engine

SkillForge Virtual Internship

Microfinance Analyst - AI Loan Eligibility Engine

Industry: Banking, Microfinance FinTech

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python / Pandas, Scikit-Learn, Data Analytics, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Stratagem Consultants**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Stratagem Consultants

Stratagem Consultants is a leading global enterprise operating at the cutting edge of the Consulting Strategy sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Stratagem Consultants Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

You have just been hired as an AI Research Analyst by a leading AI consulting firm. Your first assignment is to research a new market ripe for AI adoption. The firm is exploring opportunities to expand its AI offerings into industries that have traditionally relied on manual processes but could benefit from automation and data-driven decision-making. As part of your role, you will leverage AI tools to collect, analyze, and validate information and provide actionable recommendations to guide the firm's market entry strategy.

Conduct a detailed market research analysis using AI-powered tools. Your goal is to identify emerging trends, key players, challenges, and opportunities in the selected market. You will validate your data sources using responsible AI principles and synthesize your findings into actionable insights for decision-makers.

What Will You Learn?

- Extract key business insights from complex Consulting Strategy datasets
- Leverage Generative AI to automate exploratory data analysis
- Communicate AI-derived recommendations to executive stakeholders
- Ensure data privacy and Responsible AI compliance

What Tools Will You Learn?

- Python / Pandas
- Scikit-Learn
- Data Analytics
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Scope of Research

Scenario: Your manager has provided a vague request: "Find out how AI can transform the logistics industry." You need to refine the research scope to make it actionable. **Student Assignment:** - Break down the research question into sub-questions (e.g., "What are current pain points in logistics that AI can address?"). - Research current AI applications in logistics using Google Scholar and ChatGPT. - Identify specific areas (e.g., route optimization, inventory management) for deeper research. **Deliverable:** Submit a refined research scope document (1 page) outlining sub-questions, focus areas, and the rationale for choosing logistics as the target industry.

Task 2: Data Collection and Setup

Scenario: You need reliable data to perform your analysis. Collect information from primary (official reports, industry statistics) and secondary (blogs, AI forums) sources. **Student Assignment:** - Use AI-powered web scraping tools (or manually search) to collect recent data on logistics trends and challenges. - Organize the data into categories (e.g., pain points, solutions, emerging technologies). - Ensure sources are credible by cross-checking for bias and accuracy. **Deliverable:** Submit a structured dataset (Google Sheets or Excel) with at least 20 data points categorized and validated for your analysis.

Task 3: Build Insights Using AI Tools

Scenario: You now have raw data and need to uncover trends and actionable insights. Leverage AI tools to synthesize information. **Student Assignment:** - Use ChatGPT to summarize industry trends and challenges based on your data. - Apply NLP techniques to identify recurring terms and themes in textual data. - Highlight correlations between AI adoption and efficiency gains in logistics. **Deliverable:** Submit a concise insights report (2 pages) summarizing trends, challenges, and opportunities derived from the analysis.

Task 4: GenAI-Powered Explanation

Scenario: You need to explain complex findings to non-experts. Use generative AI tools to craft an engaging summary. **Student Assignment:** - In ChatGPT, generate a simplified explanation of your insights for stakeholders unfamiliar with AI. - Create a visual representation (infographic) that showcases key points using tools like Canva or Google Slides. **Deliverable:** Submit a one-page stakeholder-friendly summary and a visual infographic explaining your research findings.

Task 5: Audit and Responsible AI Practices

Scenario: Your firm emphasizes responsible AI use. Ensure your research adheres to ethical standards and avoids biased data. **Student Assignment:** - Review your data sources for potential bias or misinformation. - Write a short audit report detailing how you ensured ethical AI practices throughout your research. - Identify any gaps or risks in your findings. **Deliverable:** Submit a one-page Responsible AI Audit Report summarizing your validation process and ethical considerations.

Task 6: Present Your Recommendation

Scenario: The firm's leadership team needs your recommendation on whether logistics is a viable market for AI expansion. Prepare and pitch your proposal. **Student Assignment:** - Combine all your findings into a professional presentation. - Include key insights, visualizations, risks, and actionable recommendations. - Record a 5-minute video pitch summarizing your proposal. **Deliverable:** Submit a presentation deck (PDF or PPT) and a video recording of your pitch.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

3 Agentic AI Consultant - Automate a Business Process

SkillForge Virtual Internship

Agentic AI Consultant - Automate a Business Process

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, Streamlit, Process Automation, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **SynthAI Labs**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from SynthAI Labs

SynthAI Labs is a leading global enterprise operating at the cutting edge of the Tech Development sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, SynthAI Labs Insights Generative Technologies. You will gain insight into how our multi-disciplinary expert team solves some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- Streamlit
- Process Automation
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Role of Prompts

Scenario You have been hired by an enterprise that manages a large volume of customer service tickets. They want to use GenAI to classify tickets by issue type (e.g., billing, technical, feedback) and generate summaries for faster resolution.

Student Assignment Research prompt engineering fundamentals. Learn how prompt structures (context, constraints, and examples) influence GenAI outputs. Study enterprise-centric use cases, focusing on structured data generation. 8

Deliverable Submit a **1-page report** summarizing: - Key principles of prompt engineering. - How structured prompts can benefit enterprise workflows. - Examples of prompt patterns for structured outputs.

Task 2: Dataset Setup and Prompt Design

Scenario You've been provided with a sample dataset containing customer service tickets with columns like 'Ticket ID', 'Issue Description', and 'Customer Priority'. Your goal is to design initial prompts to classify these tickets by issue type and summarize them.

Student Assignment - Analyze the provided dataset structure. - Design 3 initial prompts: 1. Classify tickets into predefined categories. 2. Generate concise summaries of ticket issues. 3. Identify high-priority customer concerns.

Deliverable Submit a **document with your 3 prompts**, their intended outputs, and reasoning for their design.

Task 3: Build and Execute Prompts

Scenario Run your initial prompts on a GenAI platform (e.g., OpenAI GPT-4) and analyze the outputs. Refine your prompts iteratively to improve classification accuracy and summary quality.

Student Assignment - Test your prompts using the sample customer ticket dataset. - Adjust context, constraints, or examples to improve outputs. - Document the iterative refinements made to each prompt.

Deliverable Submit: 1. **Initial outputs** from GenAI. 2. **Refined prompts** and their improved outputs. 3. A **brief explanation** of changes made and their impact.

Task 4: Analyze Hallucinations and Explain GenAI Behavior

Scenario Some GenAI outputs contain errors (e.g., misclassifications, irrelevant summaries). Your task is to identify and explain these hallucinations, then adjust your prompts to minimize such errors.

Student Assignment - Analyze common hallucinations in the outputs (e.g., incorrect ticket classifications). - Adjust prompts to reduce errors (e.g., add constraints or improve examples). - Explain the behavior behind the hallucinations (e.g., ambiguous wording, lack of context).

Deliverable Submit a **hallucination analysis report** with: - Examples of errors. - Adjusted prompts to reduce hallucinations. - Explanations of GenAI behavior causing errors.

Task 5: Audit for Responsible AI

Scenario Ensure your prompts and outputs adhere to Responsible AI principles, including fairness,

transparency, and accuracy. Avoid biases in ticket classifications and summaries.

Student Assignment - Perform a bias audit on GenAI outputs (e.g., verify fair classification across all categories). - Document steps taken to ensure transparency and accuracy in prompts. - Suggest improvements to align with Responsible AI principles.

Deliverable Submit a **Responsible AI checklist** covering: - Bias detection results. - Transparency steps taken. - Suggestions for ethical improvements.

Task 6: Present Your Final Prompt Engineering Playbook

Scenario Compile your refined prompts, evaluation metrics, and Responsible AI findings into a **Prompt Engineering Playbook**. This will serve as a reusable resource for the enterprise.

Student Assignment - Summarize your prompt design process and key insights. - Include examples of refined prompts and outputs. - Provide actionable strategies for minimizing hallucinations and ensuring Responsible AI compliance.

Deliverable Submit your **Prompt Engineering Playbook** (5-10 pages) with: - Overview of prompt design principles. - Tested and refined prompts. - Evaluation metrics and Responsible AI strategies.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

4 Multi-Agent AI - Design an AI Workforce

SkillForge Virtual Internship

Multi-Agent AI - Design an AI Workforce

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, CrewAI / AutoGen, Prompt Engineering, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **NovaCare Health Systems**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from NovaCare Health Systems

NovaCare Health Systems is a leading global enterprise operating at the cutting edge of the Healthcare sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, NovaCare Health Systems Insights Generative Technologies. You will gain insight into how our multi – disciplinary expertssolvesome of the most complex business problems using state – of – the – art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Healthcare sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- CrewAI / AutoGen
- Prompt Engineering
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand - Analyze the Problem and Dataset

Scenario You've just joined a healthcare analytics team tasked with reducing no-shows for medical appointments. Your first step is to understand the scope of the problem and analyze the provided dataset.

Student Assignment - Analyze the impact of no-shows on healthcare efficiency (research online).
- Load the provided appointment dataset and perform an initial review of its fields, data types, 11 and missing values. - Summarize the key trends in no-show rates (e.g., by gender, age, and health condition).

Deliverable Submit a report containing: - Key insights about no-show trends (graphs and tables).
- Summary of missing values and potential data quality issues.

Task 2: Data/Setup - Preprocess and Prepare the Dataset

Scenario The dataset needs to be cleaned and prepared for predictive modeling. You'll handle missing values, encode categorical variables, and address class imbalance.

Student Assignment - Handle missing values using appropriate imputation techniques. - Encode categorical variables like 'Gender' and 'HealthCondition'. - Address class imbalance in the 'NoShow' field using techniques such as oversampling or undersampling.

Deliverable Submit a clean and preprocessed dataset ready for modeling. Include a summary of pre-processing steps taken.

Task 3: Build/Execute - Train and Evaluate the No-Show Predictor

Scenario Using the preprocessed dataset, you'll train a classification model to predict whether a patient is likely to no-show an appointment.

Student Assignment - Split the dataset into training and testing sets (e.g., 80/20 split). - Train at least two models (e.g., Logistic Regression, Random Forest). - Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.

Deliverable Submit a summary comparing the performance of different models. Include code snippets and visualizations of key metrics (e.g., confusion matrix).

Task 4: GenAI/Explanation - Interpret Predictions

Scenario Stakeholders need to understand the factors influencing no-show predictions to trust the model's recommendations. You'll use explainable AI techniques to interpret predictions.

Student Assignment - Choose one model and explain its predictions using SHAP or LIME. - Identify which features contribute most to the likelihood of a no-show (e.g., age, SMS reminder).

Deliverable Submit a document with feature importance graphs and written explanations of model predictions.

Task 5: Audit/Responsible AI - Ensure Ethical Deployment

Scenario AI models in healthcare must adhere to ethical principles. You are tasked with auditing the no-show predictor for biases.

Student Assignment - Analyze model predictions for potential biases (e.g., gender or age bias). - Propose strategies to mitigate any identified biases (e.g., reweighting samples or adding fairness constraints).

Deliverable Submit an audit report on model fairness and proposed mitigation strategies.

Task 6: Present Recommendation - Segment Patients and Propose Interventions

Scenario Based on your predictions, segment patients by no-show risk levels (e.g., high, medium, low). Develop actionable strategies to reduce no-shows for each segment.

Student Assignment - Use segmentation techniques to group patients by risk levels. - Propose intervention strategies for each segment (e.g., SMS reminders, telehealth options for high-risk patients).

Deliverable Submit a stakeholder presentation with: - Patient segmentation visualizations. - Intervention strategies tailored to each risk group.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

5 GenAI Prompt Engineer - Enterprise Prompt Lab

SkillForge Virtual Internship

GenAI Prompt Engineer - Enterprise Prompt Lab

Industry: GenAI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, Prompt Engineering, OpenAI / Claude APIs, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Cognitive Core**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Cognitive Core

Cognitive Core is a leading global enterprise operating at the cutting edge of the Agriculture sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'.

With an annual revenue exceeding 5.4billionandapresenceinover30countries, we collectivelyemployover42,000passionateprofessionals.

This jobsimulationiswithouradvancedinnovationdivision, CognitiveCoreInsightsGen – erativeTechnologies.Youwillgaininsightintohowourmulti – disciplinaryexpertssolvesomeofthe mostcomplexbusinessproblemsusingstate – of – the – artAI solutions.

The Problem:

What Will You Learn?

- Extract key business insights from complex Agriculture datasets
- Leverage Generative AI to automate exploratory data analysis
- Communicate AI-derived recommendations to executive stakeholders
- Ensure data privacy and Responsible AI compliance

What Tools Will You Learn?

- Python
- Prompt Engineering
- OpenAI / Claude APIs
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

6 Generative AI Data Analyst - Ask Data Questions with GenAI

SkillForge Virtual Internship

Generative AI Data Analyst - Ask Data Questions with GenAI

Industry: GenAI + Analytics

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python / Pandas, Data Visualization, SQL, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Global Dynamics**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Global Dynamics

Tech Global Dynamics is a leading global enterprise operating at the cutting edge of the Tech Development sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Tech Global Dynamics Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python / Pandas
- Data Visualization
- SQL
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

7 Healthcare GenAI - Clinical Knowledge Assistant

SkillForge Virtual Internship

Healthcare GenAI - Clinical Knowledge Assistant

Industry: Healthcare AI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, RAG, Vector Databases, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

Tech Innovations Ltd is a leading global enterprise operating at the cutting edge of the Tech Development sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- RAG
- Vector Databases
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

8 GenAI Cybersecurity Analyst - AI SOC Investigation

SkillForge Virtual Internship

GenAI Cybersecurity Analyst - AI SOC Investigation

Industry: Cybersecurity + AI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, Splunk / SIEM APIs, Cybersecurity Analytics, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

Tech Innovations Ltd is a leading global enterprise operating at the cutting edge of the Tech Development sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- Splunk / SIEM APIs
- Cybersecurity Analytics
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Problem - Crop Health Analysis

Scenario A group of farmers from a drought-prone region has provided crop health and environmental datasets, including satellite imagery, soil quality metrics, and weather forecasts. You need to understand the indicators of crop health and the risks involved.

Student Assignment - Research key factors affecting crop health (e.g., pests, climate, soil nutrients).
- Explore the provided dataset fields: 'Crop ID', 'Region', 'Soil Moisture', 'Temperature', 'Humidity', 'Pest Infestation', 'Yield'. - Identify relationships between environmental factors and crop health.

Deliverable Submit a summary document (PDF or Word) explaining crop health indicators and their relevance to risk prediction. Include visualizations of trends (e.g., temperature vs yield).

Task 2: Data Preparation and Setup

Scenario The provided datasets contain missing values, outliers, and redundant fields. To ensure accurate AI model predictions, you need to clean and preprocess the data.

Student Assignment - Identify and handle missing values using imputation techniques. - Remove irrelevant fields and detect outliers. - Normalize numerical fields (e.g., 'Soil Moisture', 'Temperature').

Deliverable Submit a cleaned dataset (CSV file) along with a Python notebook showcasing data preprocessing steps.

Task 3: Build the AI Model - Risk Prediction

Scenario Using the cleaned dataset, it's time to build an AI model that predicts crop health risks such as pest infestations or drought stress.

Student Assignment - Split the dataset into training and testing sets. - Implement a classification model (e.g., Decision Tree or Random Forest) to predict 'Pest Infestation' risk. - Evaluate model accuracy using metrics like precision, recall, and F1-score.

Deliverable Submit your trained model code and a report on the model's performance (e.g., confusion matrix, key metrics).

Task 4: Explainable AI - Why Did the Model Predict This?

Scenario Farmers need to understand why certain risks are predicted. You must use Explainable AI (XAI) techniques to make model predictions transparent.

Student Assignment - Use the SHAP library to explain your AI model's predictions. - Generate visual explanations for why specific predictions (e.g., pest infestation or drought stress) were made.

Deliverable Submit SHAP plots for key predictions and a brief report explaining their relevance to farmers.

Task 5: Audit for Responsible AI

Scenario AI models can unintentionally favor certain crops or regions, leading to biased outcomes. You must audit your model for fairness and ensure responsible AI practices are followed.

Student Assignment - Check for biases in predictions across regions ('Region' field). - Create a fairness report highlighting any disparities and propose solutions (e.g., retraining with balanced data).

Deliverable Submit a fairness audit report (PDF) with actionable recommendations for bias mitigation.

Task 6: Present AI Recommendations to Farmers

Scenario Based on your AI findings, prepare an advisory report for farmers. This report should include preventative measures, intervention strategies, and yield optimization techniques tailored to their crop health risks.

Student Assignment - Summarize key insights from your model's predictions. - Provide actionable recommendations (e.g., irrigation schedules, pest control techniques). - Create a visually appealing farmer advisory report.

Deliverable Submit a farmer advisory report (PDF or PowerPoint) incorporating AI- driven insights and recommendations. 21

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

9 Full Stack Developer - Build an AI Web Application

SkillForge Virtual Internship

Full Stack Developer - Build an AI Web Application

Industry: Full Stack

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Next.js / React, Node.js, OpenAI APIs, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

Tech Innovations Ltd is a leading global enterprise operating at the cutting edge of the Tech Development sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Next.js / React
- Node.js
- OpenAI APIs
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

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10 Backend Developer - Build an AI API Service

SkillForge Virtual Internship

Backend Developer - Build an AI API Service

Industry: Software

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: FastAPI / Flask, PostgreSQL, Docker / Deployment, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

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The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- FastAPI / Flask
- PostgreSQL
- Docker / Deployment
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

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Earn Certificate Post Verification

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11 AI Developer - Build a RAG Application

SkillForge Virtual Internship

AI Developer - Build a RAG Application

Industry: GenAI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangChain / LlamaIndex, Vector Databases (Pinecone), SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

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This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- LangChain / LlamaIndex
- Vector Databases (Pinecone)
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

12 LangChain Developer - Build an Enterprise AI Workflow

SkillForge Virtual Internship

LangChain Developer - Build an Enterprise AI Workflow

Industry: GenAI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangChain, Enterprise Data Integration, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

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This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- LangChain
- Enterprise Data Integration
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Company and Industry

Scenario Your firm has been approached to evaluate TechNova Inc., a mid-sized technology company specializing in AI software solutions. You must first understand the company's business model and the competitive landscape.

Student Assignment - Research TechNova's business model, revenue streams, and growth trajectory. - Identify key industry trends affecting the AI software sector. - Review TechNova's financial statements and summarize the company's current standing. 29

Deliverable A one-page summary of TechNova's business model, industry trends, and key financial metrics (e.g., revenue, EBITDA).

Task 2: Data Gathering and Setup

Scenario To perform a valuation analysis, you need to gather data for comparable companies in TechNova's sector. This includes identifying peers based on size, geography, and industry focus.

Student Assignment - Identify 5-7 comparable companies in the AI software sector using publicly available financial data. - Collect key financial metrics for these companies: revenue, EBITDA, net income, market capitalization, P/E ratio, and EV/EBITDA.

Deliverable Create an Excel spreadsheet listing the comparable companies with their financial metrics and valuation multiples.

Task 3: Build Valuation Models

Scenario With the data prepared, you will perform Comparable Company Analysis (CCA) and Precedent Transaction Analysis (PTA) to determine TechNova's valuation range.

Student Assignment - Calculate TechNova's implied valuation based on the median and mean EV/EBITDA and P/E ratios of the comparable companies. - Perform Precedent Transaction Analysis by identifying recent MA transactions in the AI sector and calculating valuation multiples.

Deliverable An Excel model showcasing valuation calculations for TechNova using both CCA and PTA methods.

Task 4: Explain Valuation Insights Using GenAI

Scenario Clients often request simplified explanations of complex models. You are tasked with interpreting your valuation results using Generative AI tools to craft a clear narrative.

Student Assignment - Use ChatGPT or similar GenAI tools to generate a client-facing explanation of TechNova's valuation. - Highlight key insights from the comparable company analysis and precedent transaction analysis.

Deliverable A 250-word client-friendly summary explaining TechNova's valuation and the rationale behind it.

Task 5: Audit and Responsible Valuation

Scenario Accuracy and integrity are critical in investment banking. You must audit your valuation for errors and ensure your recommendations align with ethical standards.

Student Assignment - Cross-check all calculations, ensuring no formula errors or data discrepancies.
- Identify any assumptions made in your analysis and validate their reasonability. - Discuss potential risks in using comparable company analysis and precedent transaction methods.

Deliverable An audit report highlighting the checks performed, assumptions validated, and risk considerations.

Task 6: Present Your Recommendations

Scenario You are presenting your findings to the firm's senior leadership team. Your Investment Memo must clearly articulate your valuation analysis and provide actionable recommendations.

Student Assignment - Compile your analysis into a professional-grade Investment Memo. - Include an executive summary, valuation details, comparison charts, and a clear recommendation for TechNova.
- Create a visually engaging presentation summarizing your findings.

Deliverable - A 3-5 page Investment Memo in PDF format. - A PowerPoint/Google Slides deck with 5-7 slides summarizing your valuation analysis and recommendations.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

13 LangGraph Developer - Build an Agentic Workflow

SkillForge Virtual Internship

LangGraph Developer - Build an Agentic Workflow

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangGraph, State Machines, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Tech Innovations Ltd.** Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Tech Innovations Ltd

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This job simulation is with our advanced innovation division, Tech Innovations Ltd Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Understand the impact of Generative AI on the Tech Development sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- LangGraph
- State Machines
- SkillForge AI Mentor

Tasks to be Performed

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

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14 AI Agent Developer - Customer Support Agent

SkillForge Virtual Internship

AI Agent Developer - Customer Support Agent

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangChain, Dialogflow / Botpress, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **CX Solutions**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from CX Solutions

CX Solutions is a leading global enterprise operating at the cutting edge of the Customer Service sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'.

With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, CX Solutions Insights Generative Technologies. You will gain insight into how our multi-disciplinary expert team solves some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

The future of customer service relies on intelligent automation. As an AI Agent Developer specializing in customer support, you are tasked with designing, training, and deploying an AI-powered virtual assistant that efficiently handles customer queries, resolves common issues, and escalates complex requests to human agents. This simulation places you in the role of a developer at a SaaS company that wants to automate FAQs and streamline escalation workflows while maintaining high customer satisfaction.

Your mission is to design and deploy a conversational AI agent capable of handling customer FAQs, integrating with support tools, escalating unresolved tickets, and ensuring seamless communication while adhering to responsible AI principles. The ultimate goal is to enhance the company's support experience and deliver measurable improvements in response time and resolution accuracy.

What Will You Learn?

- Analyze Customer Service workflows to identify Agentic AI opportunities
- Design multi-agent orchestration and human-in-the-loop handoffs
- Implement conversational AI state machines
- Validate outputs for hallucination and safety

What Tools Will You Learn?

- Python
- LangChain
- Dialogflow / Botpress
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand - Analyzing Customer Queries

Scenario: You've been provided with a dataset containing historical customer interaction logs. Your first step is to analyze the dataset, identify commonly asked questions, and categorize them into themes (e.g., billing, technical support, account management). **Student Assignment:** - Load the dataset and perform exploratory analysis to understand customer query patterns. - Identify the top 10 FAQs based on frequency. - Categorize FAQs into logical groups. **Deliverable:** Submit a report containing:

1. Visualizations showing the distribution of query types.
2. A list of top 10 FAQs and their respective categories.

Task 2: Data/Setup - Preparing the FAQ Dataset

Scenario: You need to format the FAQ data for training a conversational AI model. This involves cleaning the text, removing duplicates, and ensuring FAQs are mapped to clear response templates. **Student Assignment:** - Preprocess the FAQ dataset (remove noise, duplicates, and irrelevant data). - Structure the data into a QA format (e.g., JSON or CSV format) suitable for training. - Create sample response templates for each FAQ category. **Deliverable:** Submit the cleaned and structured FAQ dataset along with response templates.

Task 3: Build/Execute - Developing the AI Agent

Scenario: It's time to build the AI agent! Using OpenAI GPT (or an alternative NLP platform), train the model using the FAQ dataset and create dialogue flows to handle customer queries. **Student Assignment:** - Train the conversational AI model using the prepared dataset. - Build logical dialogue flows for handling different FAQ categories. - Test the model locally to ensure it responds accurately to queries. **Deliverable:** Submit the Python code (or equivalent) for training and deploying the AI model, along with screenshots of test conversations.

Task 4: GenAI/Explanation - Handling Escalations

Scenario: Not all customer queries can be resolved by the AI agent. Create an escalation workflow where unresolved queries are handed over to human agents via API integration (e.g., Zendesk or Slack). **Student Assignment:** - Develop logic to identify unresolved queries (e.g., confidence threshold or "fallback" intents). - Integrate APIs to escalate tickets to human agents. - Test the escalation workflow with sample queries. **Deliverable:** Submit the Python script for API integration and a flow diagram explaining the escalation process.

Task 5: Audit/Responsible AI - Ensuring Ethical AI Standards

Scenario: Your AI agent must comply with responsible AI principles. This includes ensuring unbiased responses, avoiding harmful outputs, and providing transparency in escalation workflows. **Student Assignment:** - Audit the AI model for bias in responses (e.g., gender, race, or cultural bias). - Implement safeguards to prevent harmful outputs. - Create a responsible AI checklist for your project. **Deliverable:** Submit the audit report, safeguards implemented, and the responsible AI checklist.

Task 6: Present Recommendation - Improving KPIs

Scenario: Present your AI agent prototype to the leadership team, emphasizing how it improves customer support KPIs like average response time, CSAT, and FCR. **Student Assignment:** - Prepare a presentation with metrics showing the agent's impact on KPIs. - Create a live demo showcasing the

agent handling queries and escalating tickets. - Recommend 35 further improvements to the AI agent.
Deliverable: Submit the presentation slides and demo video.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

Once submitted, SkillForge performs a detailed verification of your outputs. You will receive a **SkillForge Verified Virtual Internship Certificate** to share on LinkedIn.

15 AI Agent Developer - Sales Qualification Agent

SkillForge Virtual Internship

AI Agent Developer - Sales Qualification Agent

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangChain, CRM Integration (Salesforce), SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **SupportSync**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from SupportSync

SupportSync is a leading global enterprise operating at the cutting edge of the Customer Service sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'.

With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, SupportSync Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

Imagine you are part of an innovative AI startup tasked with designing a virtual receptionist for a global coworking space brand. Your goal is to create an AI-powered avatar that can welcome visitors, answer frequently asked questions, and assist with scheduling meeting rooms. The avatar must be engaging, professional, and capable of holding natural conversations with users.

In this job simulation, you will design, build, and test a conversational AI avatar. You'll start by defining a persona for the virtual receptionist, build its knowledge base, integrate conversational logic, create a script for its interactions, and finally test and improve its performance.

What Will You Learn?

- Understand the impact of Generative AI on the Customer Service sector
- Map manual processes into automated AI execution graphs
- Develop, test, and deploy AI solutions in a simulated environment
- Master prompt engineering and structured output validation

What Tools Will You Learn?

- Python
- LangChain
- CRM Integration (Salesforce)
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Problem and Define the Avatar Persona

Scenario: The coworking space brand has asked for a friendly and professional virtual receptionist that can handle interactions with users in English. The receptionist must reflect the brand's values: professionalism, inclusivity, and innovation. **Student Assignment:** - Study the company background and target audience. - Define a detailed persona for the virtual receptionist, including its appearance, tone, and key personality traits. - Draft a document that outlines the avatar's objective and scope of interactions (e.g., greeting visitors, answering FAQs, scheduling appointments). **Deliverable:** - A detailed persona document with at least the following fields: Name, Appearance, Personality Traits, Tone of Voice, and Key Responsibilities.

Task 2: Build the Knowledge Base for the Avatar

Scenario: The virtual receptionist will need to answer FAQs and provide assistance to users. You are tasked with building an initial knowledge base that will serve as the foundation for the avatar's conversational abilities. **Student Assignment:** - Research common FAQs for coworking spaces (e.g., pricing, amenities, booking rules). - Organize the questions and answers into a structured format using Google Sheets or Airtable. - Include at least 20 questions and answers in the knowledge base. **Deliverable:** - A knowledge base in Google Sheets or Airtable with fields: Question, Answer, and Tags (e.g., pricing, scheduling).

Task 3: Configure the Avatar's Chatbot and Train the AI

Scenario: It's time to bring the virtual receptionist to life! You will set up a chatbot platform, import the knowledge base, and train the AI to handle user queries. **Student Assignment:** - Choose a conversational AI platform (e.g., Dialogflow, Rasa). - Create intents and entities based on the knowledge base. - Train the AI using the imported data and refine its ability to understand user queries. **Deliverable:** - A working chatbot model that can handle the 20 FAQs with appropriate responses. - Screenshots or a short video demonstrating the chatbot's functionality.

Task 4: Generate AI Avatar Scripts for Conversations

Scenario: The virtual receptionist's conversations must feel natural and engaging. You will design scripted responses and use a generative AI tool to generate variations of these scripts. **Student Assignment:** - Write a list of 10 scripted dialogues for common scenarios (e.g., greeting visitors, answering FAQs, handling complaints). - Use a generative AI tool like OpenAI's GPT to create variations of these scripts. - Ensure the scripts align with the persona's tone and personality. **Deliverable:** - A document with the original scripts and AI-generated variations for the 10 scenarios.

Task 5: Test and Audit the Avatar for Responsible AI Practices

Scenario: Before launching the virtual receptionist, you need to test its performance and ensure it aligns with ethical AI principles, such as avoiding bias and ensuring inclusivity. **Student Assignment:** - Test the avatar with 10 different user inputs, including edge cases (e.g., ambiguous or offensive queries). - Document the avatar's responses and identify areas for improvement. - Write a short report on how the avatar adheres to responsible AI principles. **Deliverable:** - A testing report with user input, avatar output, and identified issues. - A brief summary of the ethical considerations addressed in the avatar's design.

Task 6: Present Your Final Avatar and Recommendations

Scenario: The client is excited to see the final virtual receptionist in action. You will present your work, explain your design choices, and outline recommendations for future improvements. **Student Assignment:** - Create a presentation (slides or a video) showcasing the avatar's persona, knowledge base, conversation flow, and testing results. - Include a demo of the avatar handling at least 38 5 different scenarios. - Suggest improvements or future features for the avatar. **Deliverable:** - A presentation (PDF or video) with the avatar's demo and recommendations for future iterations.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

Earn Certificate Post Verification

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16 AI Agent Developer - HR Recruitment Agent

SkillForge Virtual Internship

AI Agent Developer - HR Recruitment Agent

Industry: IT

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, LangChain, ATS Integration, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **ClientFirst Networks**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from ClientFirst Networks

ClientFirst Networks is a leading global enterprise operating at the cutting edge of the Customer Service sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4billionandapresenceinover30countries, wecollectivelyemployover42,000passionateprofessionals.

This job simulation is with our advanced innovation division, ClientFirst Networks Insights Generative Technologies. You will gain insight into how our multi – disciplinary expertssolvesomeof themostcomplexbusinessproblemsusingstate – of – the – artAI solutions.

The Problem:

Your company has collected customer health data, including engagement metrics, support tickets, product usage, and customer tenure. The leadership team has tasked you with understanding the key factors contributing to customer churn.

What Will You Learn?

- Extract key business insights from complex Customer Service datasets
- Leverage Generative AI to automate exploratory data analysis
- Communicate AI-derived recommendations to executive stakeholders
- Ensure data privacy and Responsible AI compliance

What Tools Will You Learn?

- Python
- LangChain
- ATS Integration
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand - Explore Customer Churn Dynamics

Scenario: Your company has collected customer health data, including engagement metrics, support tickets, product usage, and customer tenure. The leadership team has tasked you with understanding the key factors contributing to customer churn. 40 **Student Assignment:** - Study the dataset provided ('customer health data.csv'). - Identify and define key features (e.g., usage frequency, number of support

tickets, NPS scores). - Research and summarize the common drivers of churn in SaaS businesses. **Deliverable:** Submit a one-page summary of churn drivers with a brief description of the dataset features.

Task 2: Data/Setup - Clean and Prepare the Data

Scenario: The raw dataset has missing values and inconsistent entries. To ensure the accuracy of your analysis, you need to clean and preprocess the data. **Student Assignment:** - Handle missing or null values appropriately. - Encode categorical variables (e.g., customer segment, industry). - Scale numeric features for consistent input to machine learning models. - Split the data into training and testing sets (80/20 split). **Deliverable:** Submit a clean dataset and a description of the preprocessing steps you performed. —

Task 3: Build/Execute - Predict Churn Using Machine Learning

Scenario: You are now ready to build a predictive model to classify customers as either "churn" or "not churn." **Student Assignment:** - Train a Logistic Regression model to predict churn. - Evaluate the model's performance using metrics like accuracy, precision, recall, and F1-score. - Perform hyperparameter tuning to improve model performance. **Deliverable:** Submit the trained model, evaluation metrics, and a short explanation of how you improved model performance. —

Task 4: GenAI/Explanation - Identify Key Drivers of Churn

Scenario: Leadership wants to know *why* your model predicts certain customers are at risk of churning. Use interpretability techniques to explain the predictions. **Student Assignment:** - Use SHAP (SHapley Additive exPlanations) to identify the top features influencing churn predictions. - Visualize and summarize the contribution of each feature to the model's decisions. - Highlight one example of a high-risk customer and explain why the model flagged them. **Deliverable:** Submit a report with SHAP visualizations and an explanation of key churn drivers. —

Task 5: Audit/Responsible AI - Ensure Model Fairness

Scenario: The leadership team is concerned about potential bias in the model. For instance, are certain customer segments unfairly flagged as high risk? **Student Assignment:** - Analyze the fairness of your model across different customer demographics (e.g., industry, region). - Identify any biases and adjust the model or preprocessing steps if necessary. - Document your Responsible AI process and findings. **Deliverable:** Submit a fairness analysis report with any corrective actions taken. —

Task 6: Present Recommendation - Propose a Retention Strategy

Scenario: You've identified high-risk customers and the key drivers of churn. It's now time to present a retention strategy to the leadership team. **Student Assignment:** - Create a customer success plan for one high-risk customer, including engagement strategies, personalized recommendations, and a measurable action plan. - Summarize overall recommendations for reducing churn. **Deliverable:** Submit a 3-slide presentation or report summarizing: 1. High-risk customer profile and proposed success plan. 2. Insights from churn analysis. 3. Retention strategy recommendations.

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

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17 AI Customer Service Manager - Design a GenAI Contact Centre

SkillForge Virtual Internship

AI Customer Service Manager - Design a GenAI Contact Centre

Industry: CX / AI

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Conversational AI Design, CX Strategy, Analytics, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Marketing Global Dynamics**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Marketing Global Dynamics

Marketing Global Dynamics is a leading global enterprise operating at the cutting edge of the Marketing Branding sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4billionandapresenceinover 30countries, wecollectivelyemployover42,000passionateprofessionals.

This job simulation is with our advanced innovation division, Marketing Global Dynamics In – sights Generative Technologies. You will gain insight into how our multi – disciplinary experts solve some of the most complex business problems using state – of – the – art AI solutions.

The Problem:

What Will You Learn?

- Extract key business insights from complex Marketing Branding datasets
- Leverage Generative AI to automate exploratory data analysis
- Communicate AI-derived recommendations to executive stakeholders
- Ensure data privacy and Responsible AI compliance

What Tools Will You Learn?

- Conversational AI Design
- CX Strategy
- Analytics
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand Campaign Metrics

Scenario You are handed the campaign performance report for a recent marketing initiative. The report includes metrics like impressions, clicks, CTR, CPC, conversions, and ROAS. Your first task is to understand the KPIs and identify which campaigns are underperforming. 43

Student Assignment - Review the dataset provided, which includes metrics for three campaigns: *Google Ads*, *Facebook Ads*, and *Email Marketing*. - Calculate key performance metrics and identify trends.

Deliverable Submit a summary report highlighting: 1. Campaigns ranked by performance (e.g., highest ROI). 2. Any anomalies or trends observed.

Task 2: Data Segmentation and Audience Analysis

Scenario The marketing team suspects that poor audience targeting is contributing to underwhelming campaign performance. You are tasked with segmenting the audience based on demographics, interests, and past purchasing behavior.

Student Assignment - Use the audience data provided (age, gender, location, purchase history). - Create at least three audience segments (e.g., "Eco-conscious millennials," "Budget-friendly families").

Deliverable Submit a document with: 1. Defined audience segments and their characteristics. 2. The rationale for selecting these segments.

Task 3: Generate AI-Powered Campaign Content

Scenario Once the audience segments are defined, the marketing team needs fresh content tailored to each segment. You've been asked to use Generative AI to create ad copy for *Google Ads* and *Facebook Ads*.

Student Assignment - Use a Generative AI tool (e.g., ChatGPT) to create ad copy based on the audience segments defined in Task 2. - Ensure the content aligns with the company's eco-friendly brand voice.

Deliverable Submit: 1. Three sets of ad copy tailored to each audience segment. 2. Justification for how the content aligns with the audience's needs and the brand's identity.

Task 4: Optimise Campaigns Using AI

Scenario The marketing team has requested you to use an AI-powered optimisation tool to simulate changes to the campaign. Your goal is to increase ROI while staying within the budget.

Student Assignment - Use the provided campaign data and feed it into the AI optimisation tool (e.g., Google Ads Performance Planner). - Adjust variables like budget allocation, bid strategy, and targeting.

Deliverable Submit: 1. A before-and-after comparison of the campaign metrics (e.g., CTR, ROAS). 2. A short explanation of what changes were made and why.

Task 5: Audit for Responsible AI

Scenario Your manager is concerned about ethical considerations in using Generative AI for marketing. She asks you to assess the AI-generated content for biases and ensure it is inclusive.

Student Assignment - Review the ad copy generated in Task 3 for potential biases (e.g., stereotypes, exclusionary language). - Revise the content to ensure it aligns with ethical marketing practices.

Deliverable Submit a revised ad copy along with a brief explanation of the changes made and why they were necessary.

Task 6: Present Recommendations

Scenario You are required to present your findings and recommendations to the Marketing Director and Growth Team. This presentation is critical to securing approval for implementing your optimisation strategy.

Student Assignment - Prepare a 5-slide presentation covering: 1. Campaign performance analysis (Task 1). 2. Audience segmentation (Task 2). 3. AI-generated content (Task 3). 4. Optimisation strategy (Task 4). 5. Responsible AI audit (Task 5).

Deliverable Submit your presentation in PDF format.

MCQs Post Task Completion

1. What does CTR stand for in digital marketing? - A. Conversion Tracking Rate - B. Click-Through Rate - C. Campaign Target Reach - D. Cost-To-Revenue **Correct Answer:** B

2. What metric measures the revenue generated per dollar spent on ads? - A. CPC - B. CTR - C. ROAS - D. CPM **Correct Answer:** C

3. Which of the following is a common tool for audience segmentation? - A. Excel - B. Google Analytics - C. Canva - D. Trello **Correct Answer:** B

4. What is the primary purpose of Generative AI in marketing? - A. Email automation - B. Content creation - C. Campaign budgeting - D. Performance tracking **Correct Answer:** B

5. Which of the following is NOT a campaign performance metric? - A. Impressions - B. CTR - C. CPC - D. AI score **Correct Answer:** D

6. What does “segmentation” help marketers achieve? - A. Broader audience reach - B. Specific targeting for better campaign performance - C. Lower ad spend - D. Higher CPM **Correct Answer:** B

7. Which AI tool is commonly used for optimising Google Ads campaigns? - A. HubSpot - B. Google Ads Performance Planner - C. Canva - D. MailChimp **Correct Answer:** B

8. What is an ethical concern when using Generative AI for marketing content? - A. Content creativity - B. Biased or insensitive language - C. High ad spend - D. Low CTR **Correct Answer:** B

9. What is ROAS a measure of in digital marketing? - A. Revenue Over All Spending - B. Return On Ad Spend - C. Rate Of Audience Satisfaction - D. Reach Of Advertising Scale **Correct Answer:** B

10. Why should marketers audit AI-generated content? - A. To ensure high CTR - B. To avoid biases and ensure inclusiveness - C. To reduce CPC - D. To improve impressions **Correct Answer:** B

Submit Your Completed Task

Submission Artifacts For each task, upload: 1. **Task 1:** Campaign performance summary report. 2. **Task 2:** Audience segmentation document. 3. **Task 3:** AI-generated ad copy. 4. **Task 4:** Optimisation comparison report. 5. **Task 5:** Revised ethical ad copy and audit notes. 6. **Task 6:** Final presentation (PDF format).

Grading Rubric - Clarity of Analysis:

20

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

Submit your completed Jupyter Notebooks, analytical reports, and presentations for AI verification.

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18 Social Media Intelligence Analyst Decode a Brand Conversation

SkillForge Virtual Internship

Social Media Intelligence Analyst Decode a Brand Conversation

Industry: Marketing Branding

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python / Pandas, NLP (Spacy/NLTK), Web Scraping, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Marketing Innovations Ltd**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Marketing Innovations Ltd

Marketing Innovations Ltd is a leading global enterprise operating at the cutting edge of the Marketing Branding sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Marketing Innovations Ltd. In – sights Generative Technologies. You will gain insight into how our multi – disciplinary experts solves some of the most complex business problems using state – of – the – art AI solutions.

The Problem:

What Will You Learn?

- Extract key business insights from complex Marketing Branding datasets
- Leverage Generative AI to automate exploratory data analysis
- Communicate AI-derived recommendations to executive stakeholders
- Ensure data privacy and Responsible AI compliance

What Tools Will You Learn?

- Python / Pandas
- NLP (Spacy/NLTK)
- Web Scraping
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Brand Competitors

Scenario Your client's brand, **FitSphere**, is a direct-to-consumer fitness equipment brand. They provide high-quality, portable fitness gear for home workouts. Their key competitors are **HomeFit** and **ActiveEdge**. You need to understand the brand's market positioning and gather initial data for analysis. 46

Student Assignment 1. Research the brand FitSphere, its product offerings, and its target audience. 2. Identify 3 key competitors and compare their social media presence, including platforms they are most active on (e.g., Instagram, Twitter, TikTok). 3. Prepare a summary of FitSphere's strengths and weaknesses compared to its competitors.

Deliverable - A 2-3 slide summary of FitSphere's social media presence and its competitors' comparative analysis.

Task 2: Collect and Prepare Social Media Data

Scenario Your team has access to FitSphere's public Twitter data via an API. You need to extract 500 recent tweets mentioning the brand and prepare the data for analysis.

Student Assignment 1. Use the Twitter API (or a provided dataset) to extract 500 tweets mentioning FitSphere. 2. Clean and preprocess the data: remove duplicates, stopwords, and irrelevant tweets. 3. Organize the cleaned data in a CSV file with the following columns: - Timestamp - Username - Tweet content - Engagement metrics (likes, retweets, replies)

Deliverable - A cleaned CSV file with the required fields, ready for analysis.

Task 3: Analyze Sentiment of the Conversations

Scenario Your client wants to understand how customers feel about their brand. Use NLP-based sentiment analysis to classify the tweets as positive, negative, or neutral.

Student Assignment 1. Use Python and the VADER or TextBlob library to perform sentiment analysis on the tweets. 2. Add a new column to the CSV file for sentiment score (positive, negative, neutral). 3. Summarize the overall sentiment distribution for FitSphere.

Deliverable - Updated CSV with sentiment scores. - A bar chart or pie chart summarizing sentiment distribution.

Task 4: Identify Themes Using Generative AI

Scenario Your client wants to know the major themes in customer conversations. Summarize the key topics and sentiments using an AI-powered tool.

Student Assignment 1. Use OpenAI GPT (or another NLP model) to identify and summarize key themes in the tweets. 2. Categorize the themes into positive and negative buckets (e.g., "Quality Issues," "Great Customer Service"). 3. Write a 250-word summary highlighting the major themes.

Deliverable - A thematic analysis report with key themes and examples of tweets under each category.

Task 5: Conduct Competitor Benchmarking

Scenario Your client wants to know how their social media performance compares to competitors. Use a provided dataset containing social media metrics for FitSphere, HomeFit, and ActiveEdge.

Student Assignment 1. Analyze the dataset for key metrics such as engagement rate, sentiment distribution, and follower growth. 2. Compare FitSphere's metrics to its competitors and identify areas where the brand is leading or lagging. 3. Create a dashboard or visualizations to showcase the comparative performance.

Deliverable - A dashboard or set of visualizations comparing FitSphere's social media metrics with competitors.

Task 6: Present Recommendations

Scenario Your client expects a comprehensive report with actionable insights. They need your recommendations to improve their social media strategy based on your analysis.

Student Assignment 1. Create a 5-7 slide presentation summarizing: - Sentiment analysis and thematic insights. - Competitive benchmarking results. - Key recommendations for improving social media strategy. 2. Record a 5-minute video presenting your findings.

Deliverable - A presentation deck (PPT or PDF). - A 5-minute recorded video explaining your analysis and recommendations. — 47

Knowledge Test

Test your understanding with a knowledge assessment. You can retake it anytime.

Submit Your Completed Task

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19 UI/UX Designer Design an AI Product Experience

SkillForge Virtual Internship

UI/UX Designer Design an AI Product Experience

Industry: Design UI/UX

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Figma, User Journey Mapping, Prototyping, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Design Global Dynamics**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Design Global Dynamics

Design Global Dynamics is a leading global enterprise operating at the cutting edge of the Design UI/UX sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Design Global Dynamics Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

Imagine you've been hired as a UI/UX Designer for an innovative AI startup developing a cutting-edge product: an AI-powered mental wellness assistant.

Your goal is to ensure the product provides an intuitive, user-friendly, and visually appealing experience while addressing end-user needs effectively. Your mission is to guide the design process for this AI product by conducting user research, developing personas, designing user flows, creating wireframes, and testing usability. By the end of this simulation, you will deliver a prototype-ready design that balances functionality, accessibility, and aesthetics.

What Will You Learn?

- Map Design UI/UX business problems to AI capability matrices
- Write comprehensive Product Requirement Documents (PRDs) for AI features
- Calculate expected ROI and build a transformation roadmap
- Identify fairness, ethical, and bias implications of AI adoption

What Tools Will You Learn?

- Figma
- User Journey Mapping
- Prototyping
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand - Conduct User Research

Scenario: You are tasked with understanding the target audience for the AI mental wellness assistant. You need to uncover user pain points, preferences, and expectations for an AI-powered product that supports mental health. **Student Assignment:** - Create a Google Form survey with 10 targeted questions to gather insights on user needs (e.g., "What challenges do you face in maintaining mental wellness?"). - Conduct 3 virtual interviews with potential users to uncover qualitative insights. **Deliverable:** Submit a short report summarizing the survey results and key findings from interviews. Include charts or graphs representing quantitative data.

Task 2: Data/Setup - Develop Personas

Scenario: Based on your user research, you need to create detailed personas that represent the core segments of your audience. These personas will guide your design decisions throughout the project. **Student Assignment:** - Use Miro or a similar tool to create 3 personas. Include key details: demographics, goals, frustrations, and behaviors. - Highlight how each persona interacts with AI products and their expectations. **Deliverable:** Submit a PDF or image export of your personas from Miro, ensuring each persona is well-defined with visuals.

Task 3: Build/Execute - Design the User Flow

Scenario: You need to map out the user journey for interacting with the AI mental wellness assistant, from onboarding to using its features (e.g., mood tracking, chatbot conversations). **Student Assignment:** - Use Miro to create a user flow diagram that highlights key paths like onboarding, accessing features, and receiving insights. - Define decision points and actions at each step (e.g., "User selects mood tracker → AI provides analysis"). **Deliverable:** Submit a user flow diagram exported from Miro. Include annotations explaining each step and decision point.

Task 4: GenAI/Explanation - Create Wireframes

Scenario: It's time to convert your user flow into wireframes for core screens of the AI product. Focus on intuitive layouts and ensuring users can easily interact with AI features. **Student Assignment:** - Design wireframes for 3 primary screens: Onboarding, Mood Tracking Dashboard, and Chatbot Interface. - Ensure your designs follow accessibility guidelines (e.g., WCAG contrast standards). **Deliverable:** Submit high-quality wireframes created in Figma or Adobe XD. Include annotations describing design choices.

Task 5: Audit/Responsible AI - Conduct Usability Testing

Scenario: You need to ensure your wireframes deliver a seamless experience. Conduct usability tests to gather feedback and identify areas for improvement. **Student Assignment:** - Share your wireframes with 3 test users and ask them to perform specific tasks (e.g., "Navigate to the Mood Tracker"). - Gather feedback on usability issues and suggestions for improvement. **Deliverable:** Submit a usability test report summarizing participant feedback, including a prioritized list of design improvements.

Task 6: Present Recommendation - Final Design Proposal

Scenario: You're presenting your design recommendations to the product team. Use data from research, personas, and usability tests to justify your decisions. **Student Assignment:** - Create a presentation deck showcasing your design process: research findings, personas, user flow, wireframes, and test results. - Include actionable recommendations for the product team. **Deliverable:** Submit your presentation as a PDF or PowerPoint file. Ensure it's visually engaging and includes clear, data-driven insights.

Knowledge Test

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Submit Your Completed Task

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20 Citizen Services AI Agent Resolve a Government Complaint

SkillForge Virtual Internship

Citizen Services AI Agent Resolve a Government Complaint

Industry: Government Public Services

Duration: 3-4 hours

Mode: Individual, task-based job simulation

Primary Skills: Python, Data Privacy Frameworks, NLP, SkillForge AI Mentor

Complete work that simulates life on the job. 3-4 hours and self-paced.
Stand out in your application to **Government Global Dynamics**. Show us you're a motivated candidate.

Overview

What is this Job Simulation All About?

Introduction from Government Global Dynamics

Government Global Dynamics is a leading global enterprise operating at the cutting edge of the Government Public Services sector, with a mission 'To revolutionize service delivery and create long-term stakeholder value through trusted innovation'. With an annual revenue exceeding 5.4 billion and a presence in over 30 countries, we collectively employ over 42,000 passionate professionals.

This job simulation is with our advanced innovation division, Government Global Dynamics Insights Generative Technologies. You will gain insight into how our multi-disciplinary experts solve some of the most complex business problems using state-of-the-art AI solutions.

The Problem:

What Will You Learn?

- Analyze Government Public Services workflows to identify Agentic AI opportunities
- Design multi-agent orchestration and human-in-the-loop handoffs
- Implement conversational AI state machines
- Validate outputs for hallucination and safety

What Tools Will You Learn?

- Python
- Data Privacy Frameworks
- NLP
- SkillForge AI Mentor

Tasks to be Performed

Task 1: Understand the Problem

Scenario A government services portal receives 10,000+ citizen complaints every month. These complaints vary from potholes and waste management to utility bill disputes. Your first step is to understand the dataset and problem statement. 52

Student Assignment 1. Explore a provided dataset containing complaints with fields: - ‘Complaint ID’: Unique identifier. - ‘Citizen Name’: Name of complainant. - ‘Complaint Text’: The full description of the complaint. - ‘Department’: Department responsible for resolution (e.g., Waste Management, Public Works). - ‘Urgency’: Low, Medium, High. - ‘Status’: Resolved, Pending, Escalated. 2. Analyze the dataset for patterns, such as the most common complaint types and departments with the highest complaint volumes.

Deliverable Submit a short report (300 words) with: - Key patterns in the dataset. - Suggestions for how AI can reduce backlog and improve response times.

Task 2: Prepare the Data

Scenario You must clean and preprocess the complaint data to prepare it for text classification.

Student Assignment 1. Perform text preprocessing on the ‘Complaint Text’ field, including: - Tokenization. - Stopword removal. - Lemmatization. 2. Split the dataset into training, validation, and test sets. 3. Encode the labels (‘Department’) for classification using one-hot encoding or label encoding.

Deliverable Submit the cleaned dataset and a summary of your preprocessing steps.

Task 3: Build the Classification Model

Scenario Now, you will build a machine learning model to classify complaints into the correct department.

Student Assignment 1. Train a text classification model using Hugging Face Transformers (e.g., DistilBERT). 2. Evaluate the model using metrics like accuracy, precision, recall, and F1-score. Aim for an F1-score ≥ 0.8 . 3. Save the trained model for later use in the pipeline.

Deliverable Submit the trained model file and a report on model performance (include a confusion matrix).

Task 4: Integrate Knowledge Retrieval

Scenario Once a complaint is classified, the AI Agent should retrieve relevant knowledge articles to suggest resolutions.

Student Assignment 1. Load a provided knowledge base with fields: - ‘Article ID’: Unique identifier. - ‘Department’: Associated department. - ‘Keywords’: Tags for retrieval. - ‘Article Text’: Full text of the article. 2. Use embeddings (e.g., SentenceTransformers) to generate vector representations for the articles and complaints. 3. Implement a similarity search algorithm (e.g., cosine similarity) to retrieve the top 3 articles for each complaint.

Deliverable Submit your knowledge retrieval script and an output file showing the top 3 articles for 10 test complaints.

Task 5: Test Responsible AI Practices

Scenario AI systems for citizen services must be fair and unbiased. You need to audit your classification model for potential biases.

Student Assignment 1. Analyze model performance across different demographics (e.g., names indicating gender or region). 2. Check for overrepresentation or neglect of certain complaint categories. 3. Propose methods to address any bias found (e.g., rebalancing training data).

Deliverable Submit an audit report (300 words) outlining your findings and proposed solutions.

Task 6: Present Final Recommendation

Scenario You are pitching your solution to government stakeholders. Summarize your AI pipeline and demonstrate its impact on complaint resolution efficiency.

Student Assignment 1. Create a presentation (5 slides) covering: - Problem statement and dataset analysis. - Classification model results. - Knowledge retrieval accuracy. - Responsible AI findings. - Overall impact and next steps. 2. Record a 3-minute video walkthrough of your presentation.

Deliverable Submit your slides and video walkthrough.

Knowledge Test

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